# Gen AI-Enabled Werewolf Game: A Prototype for Chinese Undergraduates’ English Oral Practice

*EDS 6019: AI for Teaching and Learning Group Project*


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## 1. Problem Statement and Significance

## Problem Statement

1. The test-oriented English education in mainland China focuses more on test-taking skills (such as rote memorization of grammar, repeated practice of test questions, and mechanical vocabulary training) rather than practicing English in real life.

2. Students who practice English in this mode suffer from a writing-but-not-speaking English pattern, which reflects poor oral English skills and a tendency towards Chinese English and an inability to truly interact in spoken English.

3. This type of exam-oriented education in mainland China does not allow students to use English to access diverse global information and participate in international dialogue, which weakens the role of English as a communication tool and cultural background.

4. This approach prevents English education from fulfilling its essential role as a practical communication tool and a "humanistic role" and as a cultural bond. “instrumental function” and “humanistic role.”

## Significance

As a global language, English matters crucially. First, it serves as a communication tool and cultural bridge. Second, the long-standing exam-oriented education model deprives students of using English to access diverse information and join global conversations—undermining their competitiveness in the global workforce, deviating from fostering international perspectives and cross-cultural competence, and hindering English education from fully fulfilling its "instrumental" and "humanistic" functions.

## 2. Goals and Objectives

## Core Goal

Develop a functionally demonstrable Gen AI-enabled Werewolf game prototype tailored for Chinese undergraduates, using immersive social gaming dynamics to drive tangible improvements in their English oral fluency and communicative confidence.

## Specific Measurable Objectives

1. **Oral Expression Extension**: Post-prototype launch, target users (non-English majors) will speak English for an average of 3 minutes or longer per round—a 50% increase compared to traditional oral practice tools.
2. **Grammar Precision Advancement**: After 4 weeks of consistent use, the rate of grammatical errors in users' spoken English will drop by 30%, with metrics quantified and validated via the prototype's AI feedback mechanism.
3. **Dual-Dimension Satisfaction**: User satisfaction scores for the prototype's "gameplay immersion" and "AI feedback value" will each reach 4.0 or higher on a 5-point Likert scale, confirming its balance of entertainment and language-learning efficacy.

# 3. Target Audience and Context

1. **Intended Users**: Chinese non-English major undergraduates seeking to improve oral English. They may need practice for CET-4/CET-6 exams or workplace communication but lack access to real-person practice opportunities.
2. **Learning Environment**: Deployed in fragmented post-class scenarios (e.g., evening self-study, commuting, weekends). Learners practice independently or engage in online multiplayer sessions via mobile phones or computers, with no fixed venue—adaptable to college-friendly spaces like libraries, dorms, homes, and classrooms.

# 4. Review of Existing Solutions

Current tools fail to meet the dual demands of "immersion + instant feedback," with key limitations:

1. **Offline English Corners**: Restricted by time and location (e.g., students in remote areas or with busy schedules struggle to attend regularly). In large groups, individual speaking time is ≤1 minute, and personalized feedback is absent, leading to uncorrected repeated mistakes.
2. **Tutoring Apps (e.g., Cambly)**: High cost (≈¥50 per session) and scheduling challenges; students often speak less to "maximize tutor time."
3. **Language Learning Apps (e.g., Duolingo)**: Overfocus on vocabulary/grammar drills; "fill-in-the-blank" exercises do not simulate the spontaneity of real conversations.
4. **Dedicated Werewolf Apps (e.g., NetEase Werewolf)**: Designed solely for entertainment, with no language-learning components or feedback on grammar/logical flow.

## Our Improvement

This solution leverages Werewolf’s social dynamics to motivate natural English expression, combined with Gen AI for speech-to-text conversion, grammar refinement, and contextual feedback—effectively balancing engagement and educational value.

# 5. Proposed GenAI Solution

## Concept

From a user perspective, the prototype’s workflow is as follows:

1. **Access & Selection**: Users enter via a Replit project link, then choose difficulty (basic: simple sentences; medium: complex expressions; advanced: logical debate) and role (e.g., werewolf, prophet, villager).
2. **Game Launch**: The system matches 10–12 online players into a room; an AI host (“judge”) explains rules via voice/text (e.g., “Werewolves must kill one player nightly”).
3. **In-Game Communication**: Users use speech-to-text (e.g., “Player 6 is suspicious—he avoided his night action”); the prototype auto-translates speech to text and shares it in the room.
4. **AI Feedback**: Post-speech, AI provides revisions (e.g., “I very like your thinking” → “I like your opinion very much”) and logical prompts (e.g., “Explain why you suspect Player 6”).
5. **Post-Game Report**: Users receive feedback on grammar errors, context-specific vocabulary (e.g., “I am sure” → “there’s no doubt”), fluency, and debate logic (e.g., “Add ‘viewpoint + basis’ to 4/5 speeches: ‘The reason I think… is that…’”).

## Technology

6. **Core Platform: Replit**
    1. No coding background required—uses natural language instructions, templates, and a visual editor to support team collaboration on component development.
    2. Easy API integration (e.g., OpenAI, speech tools) without extra server configuration, enabling full-process functionality.
    3. Demo-friendly: Generates permanent web links for multi-device access, facilitating class presentations, video recording, and testing.
7. **Auxiliary Tool**: ChatGPT, used to translate instructions into professional technical language.

## Implementation Sketch

1. **Design (10.6–10.12)**: 2 members handle speech-to-text and UI design; 2 manage game hosting/flow; 1 researches oral English feedback frameworks for in-game text.

2. **Development (10.13–10.26)**: Build the basic game flow (“enter room → host broadcasts rules → user speech-to-text”); test links for smooth operation; collect AI feedback data.
3. **Testing & Improvement (10.27–11.5)**: Recruit 5 undergraduates to test speech-to-text accuracy and interaction fluency; fix lag and recognition errors.
4. **Finalization (11.6–11.12)**: Save the prototype, generate a share link, and record a “login → speak → feedback” video for YouTube presentations.